

CFD Investigation of a Coupled Axial-Compressor Rotor–Stator Stage

Sequential cascade baseline and coupled thin-annulus
frozen-rotor MRF extension

OpenFOAM v2412 – steady incompressible RANS – $k-\omega$ SST

Author: Alexandre Sigiridis-Schlepp
Project type: Mechanical/CFD engineering portfolio project
Geometry: periodic thin-annulus rotor–stator stage
Solver: `simpleFoam`
Software: OpenFOAM v2412, Gmsh, Python, ParaView

This project is intentionally a reduced-order educational compressor-stage model. The final extension couples one rotor passage and one stator passage in a periodic thin annulus using a steady MRF/frozen-rotor treatment. It is not a full-annulus transient compressor simulation and does not claim to predict rotating stall.

Contents

1	Project objective	2
2	Baseline physical model	2
2.1	Unwrapped blade-to-blade representation	2
2.2	Absolute, blade and relative velocities	3
2.3	Rotor turning and Euler work	4
3	Blade geometry	5
4	Mesh generation	5
5	Flow model and boundary conditions	6
5.1	Governing equations	6
5.2	Turbulence model	7
5.3	Boundary conditions	8
6	Baseline sequential rotor–stator workflow	8
6.1	Rotor calculation	8
6.2	Stator calculation	9
7	Off-design incidence sweep	10
8	Pressure losses	11
9	Baseline sequential visualization	12
10	Current quantitative status	13
11	Coupled rotor–stator MRF stage	13
11.1	Thin-annulus geometry and periodicity	13
11.2	Coupled mesh	14
11.3	MRF formulation	14
11.4	Numerical setup and staged start-up	15
11.5	Convergence records	16
11.6	Coupled-stage post-processing definitions	17
11.7	Station-averaged exported values	17
11.8	Exported stage metrics	18
11.9	Post-processing figures	19
11.10	Full-annulus visualization	19
12	Limitations of the coupled model	20
13	Next numerical extension	21
14	Conclusion	21

1 Project objective

The purpose of this project is to reproduce, with a lightweight and fully automated CFD workflow, several fundamental aerodynamic mechanisms of an axial compressor:

1. the velocity triangle seen by a rotor blade;
2. turning of the flow through a cambered rotor cascade;
3. reconstruction of the absolute rotor-exit velocity from the relative CFD solution;
4. removal of tangential velocity by a downstream stator cascade;
5. change of rotor incidence when the axial velocity is reduced at constant blade speed;
6. the associated development of off-design separation.

The project deliberately avoids the complexity of a complete turbomachine. Only one blade passage is solved for each row. Periodicity then represents an infinite linear cascade. This keeps the model small enough to remain transparent while retaining the main velocity-triangle physics.

The baseline workflow is *sequential*: the rotor is solved first, its mean exit velocity is extracted, converted back to the absolute frame, and then used as the inlet condition for the stator. Thus the baseline model is one-way coupled,

$$\text{rotor} \longrightarrow V_2 \longrightarrow \text{stator},$$

rather than a fully coupled rotor–stator stage.

A second stage of the project then replaces this one-way workflow with a single coupled rotor–stator domain. The rotor and stator are solved simultaneously in `simpleFoam`; a cell-zone MRF treatment is applied to the rotor region while the stator remains stationary. This coupled model is documented in Sec. 11.

2 Baseline physical model

2.1 Unwrapped blade-to-blade representation

At a fixed mean radius of an axial compressor, the cylindrical annulus is locally unwrapped into the Cartesian plane

$$x = \text{axial direction}, \quad y = \text{tangential direction}, \quad z = \text{radial direction}.$$

The flow is assumed invariant in the radial direction. OpenFOAM nevertheless requires a volume mesh, so the 2D geometry is extruded through one cell in z . The two radial faces are assigned the `empty` boundary condition, which enforces a two-dimensional solution.

The pitch is

$$s = 50 \text{ mm},$$

and the chord is

$$c = 60 \text{ mm},$$

giving a cascade solidity

$$\sigma = \frac{c}{s} = 1.2.$$

Top and bottom boundaries are cyclic. Therefore one numerical passage represents the periodically repeated cascade.

2.2 Absolute, blade and relative velocities

Three velocities must be distinguished carefully:

$\mathbf{V}$ = absolute fluid velocity,

$\mathbf{U}_b$ = blade peripheral velocity,

$\mathbf{W}$ = velocity relative to the moving blade.

Their relationship is

$$\boxed{\mathbf{W} = \mathbf{V} - \mathbf{U}_b} \quad (1)$$

or equivalently

$$\boxed{\mathbf{V} = \mathbf{W} + \mathbf{U}_b}. \quad (2)$$

At the design point the absolute inlet velocity is purely axial,

$$\mathbf{V}_1 = \begin{pmatrix} 60 \\ 0 \\ 0 \end{pmatrix} \text{ m/s},$$

while the prescribed local blade speed is

$$\mathbf{U}_b = \begin{pmatrix} 0 \\ 80 \\ 0 \end{pmatrix} \text{ m/s}.$$

The rotor therefore sees

$$\mathbf{W}_1 = \mathbf{V}_1 - \mathbf{U}_b = \begin{pmatrix} 60 \\ -80 \\ 0 \end{pmatrix} \text{ m/s},$$

with

$$|\mathbf{W}_1| = \sqrt{60^2 + 80^2} = 100 \text{ m/s}.$$

The inlet relative flow angle is

$$\beta_1 = \text{atan2}(W_{1y}, W_{1x}) = \text{atan2}(-80, 60) = -53.13^\circ. \quad (3)$$

The rotor leading-edge metal angle is approximately

$$\beta_{\text{LE}} = -53^\circ,$$

so the design incidence,

$$i = \beta_{\text{flow}} - \beta_{\text{LE}}, \quad (4)$$

is close to zero:

$$i_{\text{design}} \approx -0.13^\circ.$$

This is the intended design condition.

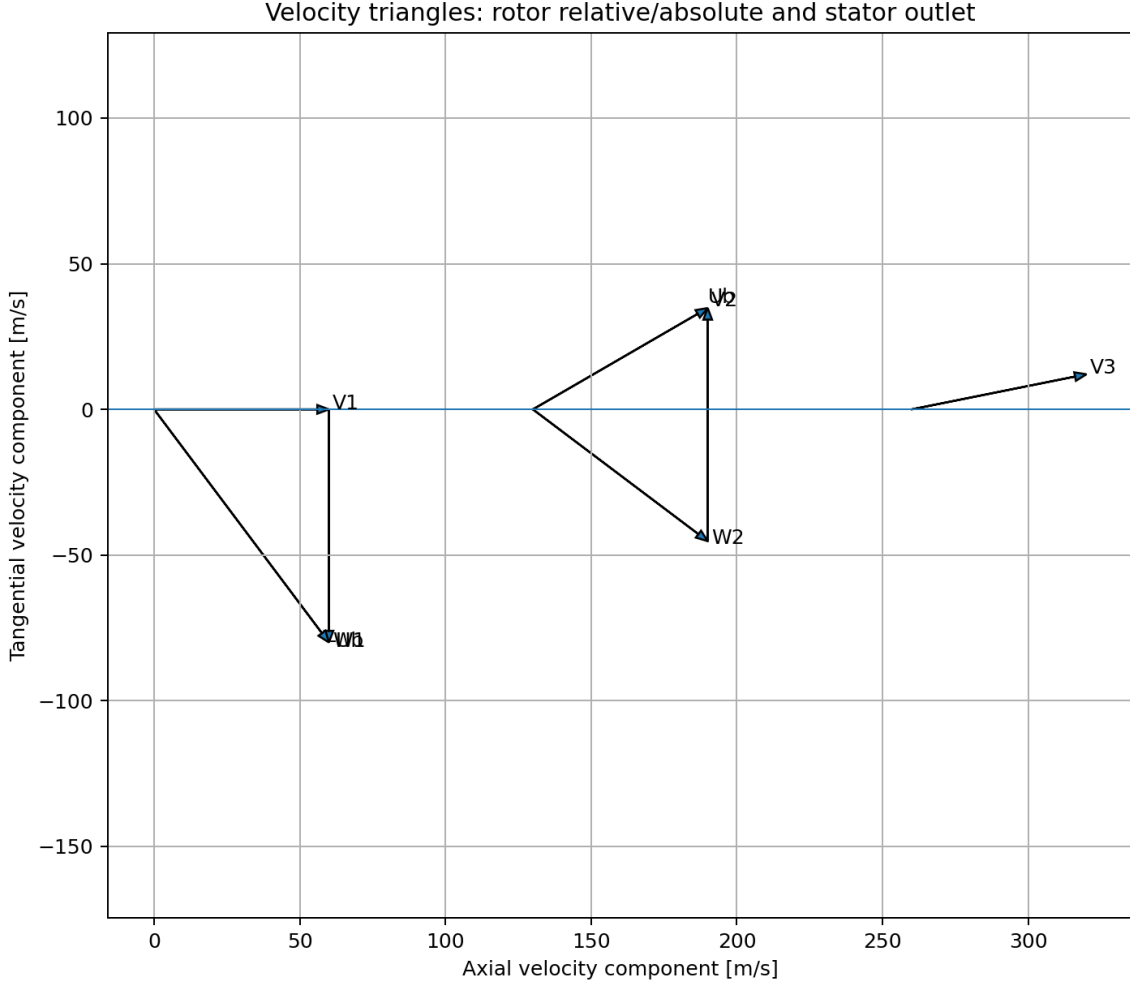


Figure 1: Velocity-triangle representation used by the project. The rotor is solved using the relative velocity $\mathbf{W}$, whereas the stator is solved using the absolute velocity $\mathbf{V}$.

2.3 Rotor turning and Euler work

The rotor changes the tangential component of the absolute flow velocity. After the relative rotor-exit velocity $\mathbf{W}_2$ is obtained from CFD, the absolute velocity is reconstructed with Eq. (2):

$$V_{2x} = W_{2x}, \quad V_{2y} = W_{2y} + U_b.$$

The corresponding absolute flow angle is

$$\alpha_2 = \text{atan2}(V_{2y}, V_{2x}). \quad (5)$$

For a real axial compressor, the Euler turbomachinery equation gives the specific stagnation-enthalpy exchange

$$\Delta h_0 = U_b \Delta V_{\theta}. \quad (6)$$

Since the design inlet has $V_{\theta 1} = 0$,

$$\Delta h_{0,\text{Euler}} = U_b V_{\theta 2}. \quad (7)$$

In this project Eq. (7) is used only as a pedagogical estimate derived from the velocity triangle. The incompressible 2D cascade is not used to claim a real compressor-stage pressure ratio.

3 Blade geometry

A symmetric NACA profile rotated to an arbitrary angle would not adequately represent a compressor blade. Both rotor and stator are therefore generated from a prescribed camber line.

The camber line is a cubic Hermite curve between the leading and trailing edges. If $t \in [0, 1]$ is the curve parameter,

$$\mathbf{C}(t) = h_{00}\mathbf{P}_0 + h_{10}\mathbf{m}_0 + h_{01}\mathbf{P}_1 + h_{11}\mathbf{m}_1, \quad (8)$$

with the usual Hermite basis functions

$$h_{00} = 2t^3 - 3t^2 + 1, \quad h_{10} = t^3 - 2t^2 + t,$$

$$h_{01} = -2t^3 + 3t^2, \quad h_{11} = t^3 - t^2.$$

The endpoint tangent vectors are chosen from the desired metal angles. For the rotor,

$$\beta_{\text{LE}} \approx -53^\circ, \quad \beta_{\text{TE}} \approx -28^\circ,$$

with a chord stagger of approximately

$$\gamma = -40^\circ.$$

A NACA-like thickness distribution is then applied normal to the local camber line,

$$y_t = 5t_c c (0.2969\sqrt{t} - 0.1260t - 0.3516t^2 + 0.2843t^3 - 0.1036t^4), \quad (9)$$

with a maximum thickness close to 10% of the chord. A small finite trailing-edge thickness is retained to make the geometry and mesh robust.

For the nominal rotor, the generated domain extends approximately over

$$x \in [-0.1134, 0.1731] \text{ m}, \quad y \in [-0.025, 0.025] \text{ m}.$$

4 Mesh generation

Gmsh is used because it allows the geometry, periodicity, local refinement and boundary-layer mesh to be generated parametrically from Python.

The mesh strategy combines:

- approximately 1.10 mm far-field sizing;
- approximately 0.35 mm sizing near the blade;
- a first boundary-layer spacing of about 0.07 mm;
- a total boundary-layer thickness of about 1.4 mm;
- a boundary-layer growth ratio of 1.22;
- additional wake refinement downstream of the trailing edge.

The 2D mesh is extruded through a single 1 mm layer. The radial faces are then converted to empty patches.

The nominal rotor mesh obtained during execution contained

$$N_{\text{cells}} = 87\,053.$$

The periodic top and bottom boundaries each contained

$$N_{\text{periodic}} = 412$$

faces, and the automated mesh validation returned **Mesh OK**. The reported maximum skewness was

$$\text{skew}_{\text{max}} = 2.125.$$

The non-orthogonality values were not recovered by the first version of the mesh-statistics parser even though **checkMesh** passed; they should therefore be read directly from the final **checkMesh** log before publication rather than guessed.

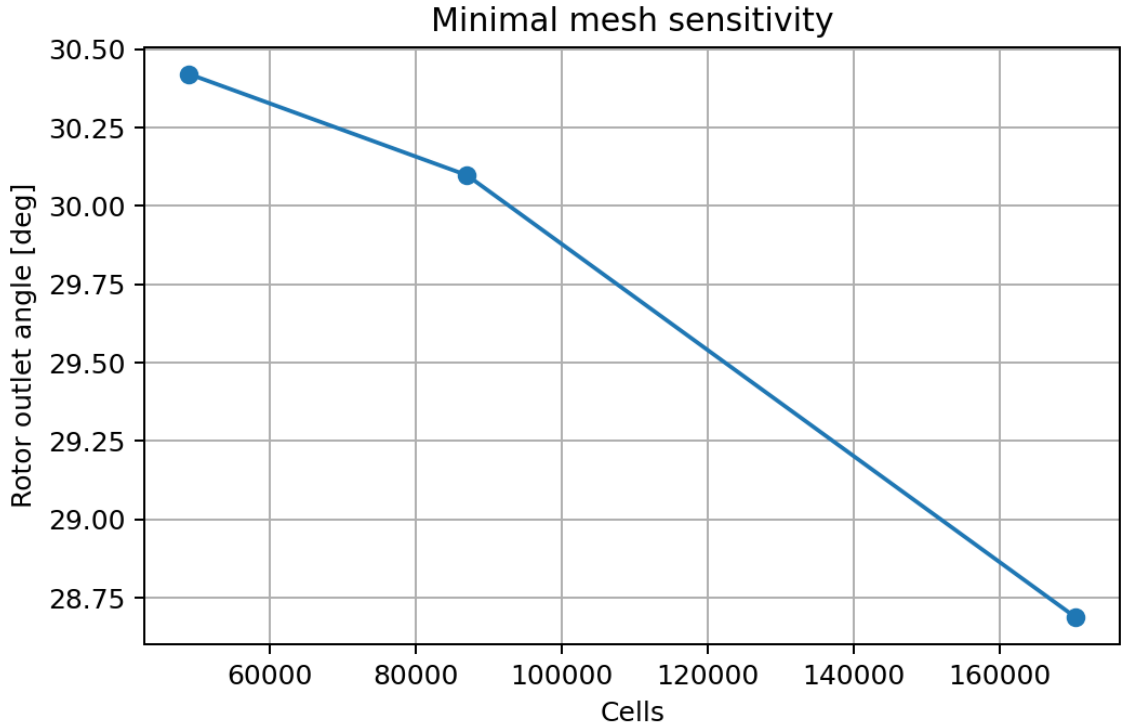


Figure 2: Minimal mesh-sensitivity study. Coarse, nominal and fine meshes are compared using outlet angle, flow rate and total-pressure-loss quantities.

5 Flow model and boundary conditions

5.1 Governing equations

The fluid is modelled as steady, incompressible air with

$$\rho = 1.225 \text{ kg/m}^3, \quad \nu = 1.5 \times 10^{-5} \text{ m}^2/\text{s}.$$

The Reynolds-averaged continuity and momentum equations are

$$\nabla \cdot \mathbf{U} = 0, \tag{10}$$

$$\rho(\mathbf{U} \cdot \nabla)\mathbf{U} = -\nabla p + \nabla \cdot [(\mu + \mu_t)(\nabla\mathbf{U} + \nabla\mathbf{U}^T)]. \quad (11)$$

The solver is `simpleFoam`, which uses the SIMPLE algorithm for the pressure–velocity coupling.

In the rotor cases, the OpenFOAM field $\mathbf{U}$ represents the relative velocity $\mathbf{W}$. In the stator case, $\mathbf{U}$ represents the absolute velocity $\mathbf{V}$.

The design relative Reynolds number based on chord is approximately

$$Re_c = \frac{|\mathbf{W}_1|c}{\nu} = \frac{100 \times 0.06}{1.5 \times 10^{-5}} \approx 4.0 \times 10^5. \quad (12)$$

Using a representative speed of sound of approximately 343 m/s, the maximum design inlet relative Mach number is roughly

$$M_{\text{rel}} \approx \frac{100}{343} \approx 0.29.$$

The incompressible assumption is therefore close to the usual $M \lesssim 0.3$ engineering threshold and is acceptable for the intended pedagogical study, although compressibility would become relevant if the characteristic velocity were increased.

5.2 Turbulence model

The turbulence closure is the k – ω SST model. It is a useful choice for turbomachinery-type external and internal flows because it combines a near-wall k – ω treatment with a k – ε -like behaviour away from the wall and includes a shear-stress limiter designed to improve separation prediction.

The inlet turbulence intensity is

$$I = 4\%$$

and the turbulent length scale is taken as

$$\ell_t = 0.07c = 4.2 \text{ mm}.$$

The inlet turbulent kinetic energy is therefore

$$k = \frac{3}{2}(I|\mathbf{U}|)^2. \quad (13)$$

For $|\mathbf{W}_1| = 100 \text{ m/s}$,

$$k = \frac{3}{2}(0.04 \times 100)^2 = \boxed{24 \text{ m}^2/\text{s}^2}.$$

Using

$$\omega = \frac{\sqrt{k}}{C_\mu^{1/4}\ell_t}, \quad C_\mu = 0.09, \quad (14)$$

gives

$$\omega \approx \boxed{2.13 \times 10^3/\text{s}}.$$

The blade uses wall functions compatible with the SST setup: `kqRWallFunction`, `omegaWallFunction` and `nutkWallFunction`. The project does not claim a strict low-Reynolds wall-resolved $y^+ \approx 1$ calculation.

5.3 Boundary conditions

The main boundary conditions are:

Table 1: Main boundary conditions.

Boundary	Velocity	Pressure
Inlet	fixed value	zero gradient
Outlet	<code>pressureInletOutletVelocity</code>	fixed reference $p = 0$
Blade	no slip	zero gradient
Top/bottom	<code>cyclic</code>	<code>cyclic</code>
Front/back	<code>empty</code>	<code>empty</code>

For incompressible OpenFOAM, the pressure field used by `simpleFoam` is kinematic pressure,

$$p_{\text{OF}} = \frac{p_{\text{phys}}}{\rho}.$$

Consequently, setting $p = 0$ at the outlet merely establishes the pressure reference; it does not represent zero absolute pressure.

6 Baseline sequential rotor–stator workflow

6.1 Rotor calculation

The rotor blade is held fixed numerically and is supplied with the relative inlet velocity

$$\mathbf{W}_1 = (60, -80, 0) \text{ m/s}.$$

No outlet flow angle is imposed. The CFD solution must determine the rotor turning itself. The final solver log of the nominal rotor showed the following last equation-residual records:

Table 2: Last recorded residuals for the nominal rotor calculation.

Variable	Initial residual	Final linear-solver residual
U_x	1.346×10^{-8}	2.828×10^{-9}
U_y	6.994×10^{-9}	6.994×10^{-9}
p	1.013×10^{-8}	5.173×10^{-9}
k	9.891×10^{-7}	1.866×10^{-8}
ω	9.960×10^{-9}	9.960×10^{-9}

These values are below the SIMPLE convergence criteria used for the case. Residual convergence alone, however, is not considered sufficient: mass balance, mesh quality and the physical flow field must also be inspected.

After convergence, the mean rotor-exit velocity $\mathbf{W}_2$ is extracted and the absolute velocity is reconstructed,

$$\mathbf{V}_2 = \mathbf{W}_2 + \mathbf{U}_b.$$

The post-processing script also creates a `Vabs` field to avoid confusing relative and absolute velocity in ParaView.

6.2 Stator calculation

The stator is solved as a second steady cascade. Its inlet condition is not chosen arbitrarily: it is generated from the actual rotor-exit absolute velocity $\mathbf{V}_2$.

The stator geometry is also adapted to that inlet angle. Its leading-edge metal angle is initialized close to

$$\beta_{\text{LE,stator}} \approx \alpha_2 - 2^\circ,$$

while the trailing-edge angle is close to the axial direction,

$$\beta_{\text{TE,stator}} \approx 3^\circ.$$

The intended aerodynamic effect is

$$|V_{\theta 3}| < |V_{\theta 2}|$$

and therefore

$$|\alpha_3| < |\alpha_2|.$$

A useful straightening indicator is

$$\eta_{\text{straight}} = 1 - \frac{|\alpha_3|}{|\alpha_2|}. \quad (15)$$

The corresponding velocity-component indicator is

$$\eta_\theta = 1 - \frac{|V_{\theta 3}|}{|V_{\theta 2}|}. \quad (16)$$

The actual values should be copied directly from `results/summary.csv`; they are intentionally not fabricated in this document.

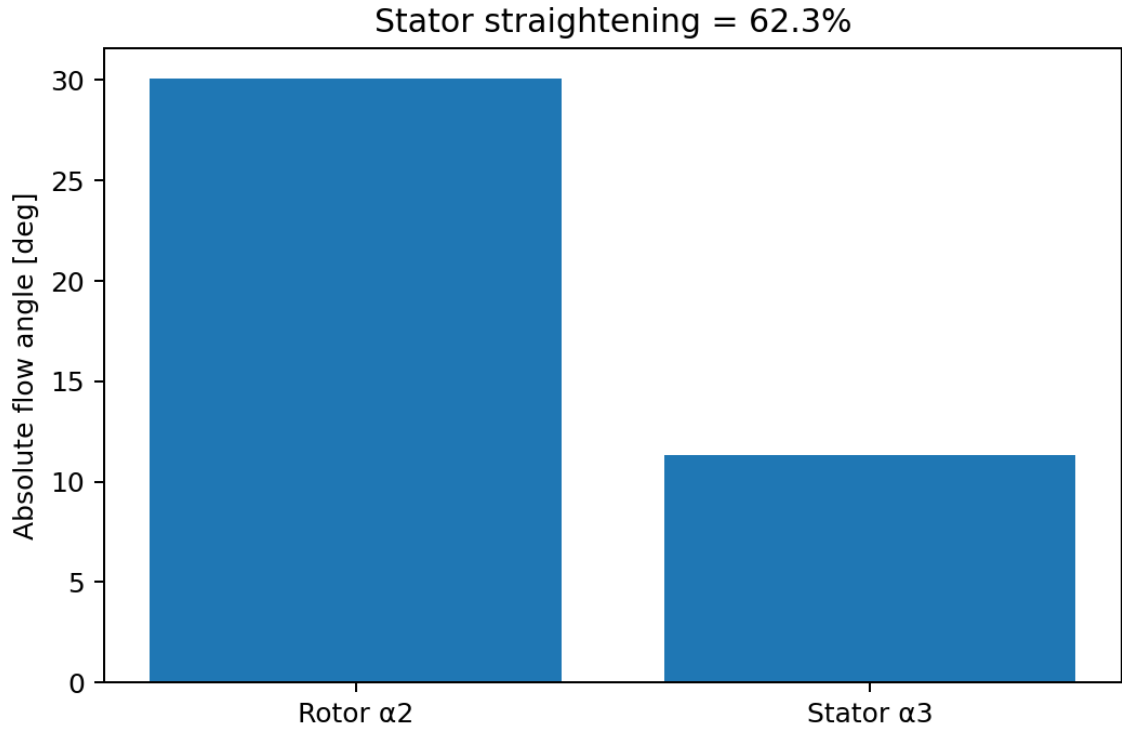


Figure 3: Absolute rotor-exit and stator-exit flow angles. The expected trend is a positive rotor-exit swirl followed by substantial stator straightening.

7 Off-design incidence sweep

The rotor geometry and blade speed remain fixed while the absolute axial inlet velocity is reduced. The relative inlet velocity becomes

$$\mathbf{W}_{\text{in}} = (V_x, -80, 0).$$

Therefore,

$$\beta_{\text{flow}} = \text{atan2}(-80, V_x). \quad (17)$$

The change in velocity triangle alone modifies incidence. For the current rotor leading-edge metal angle $\beta_{\text{LE}} = -53^\circ$:

Table 3: Theoretical inlet-angle change across the off-design sweep.

V_x [m/s]	β_{flow} [deg]	$i = \beta_{\text{flow}} - \beta_{\text{LE}}$ [deg]
70	-48.81	+4.19
60	-53.13	-0.13
50	-57.99	-4.99
40	-63.43	-10.43
35	-66.37	-13.37

The important point is that no blade geometry is changed. The off-design behaviour is caused by the changing velocity triangle. As V_x decreases, the magnitude of the incidence departure from the design condition increases, which raises the adverse pressure-gradient burden on the blade and can drive boundary-layer separation.

For a boundary layer to remain attached, near-wall fluid must retain enough streamwise momentum to overcome the adverse pressure gradient. In a qualitative integral sense, increasing incidence strengthens the suction-side loading and can make

$$\frac{\partial p}{\partial s} > 0$$

large enough that the wall shear stress approaches zero,

$$\tau_w = \mu \left. \frac{\partial u_t}{\partial n} \right|_w \rightarrow 0.$$

A change of sign of τ_w is associated with local reversed flow and separation.

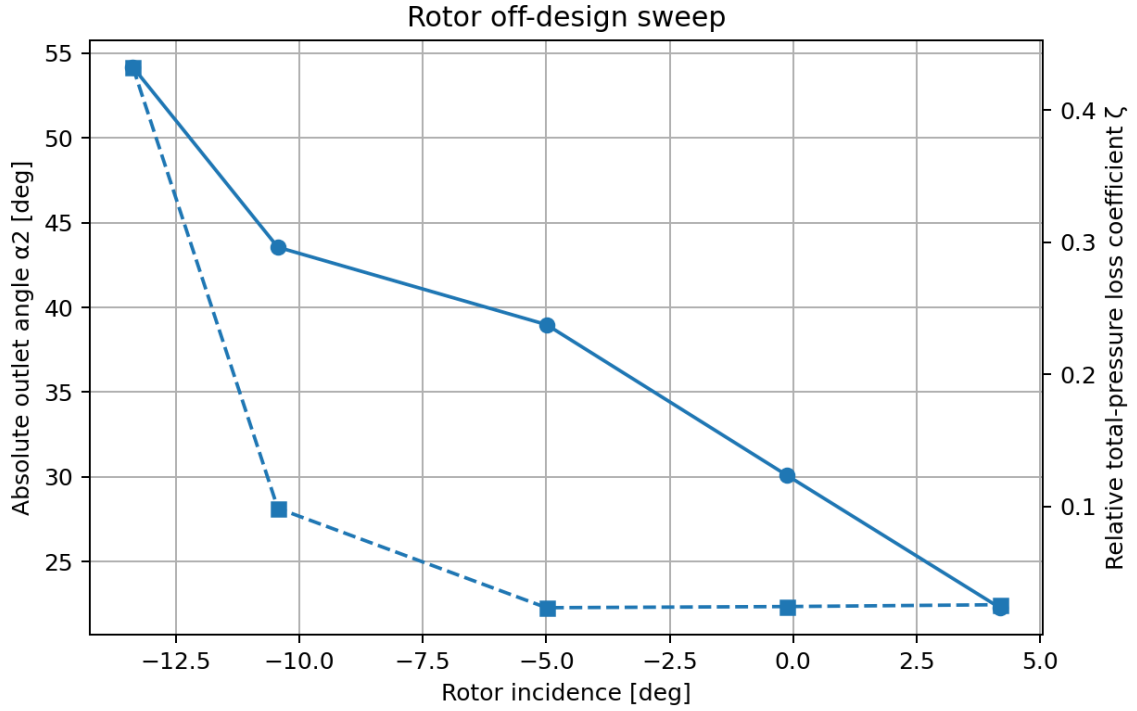


Figure 4: Rotor off-design sweep. The plot should be interpreted together with the ParaView fields: a numerical change in loss or outlet angle is not, by itself, proof of separation.

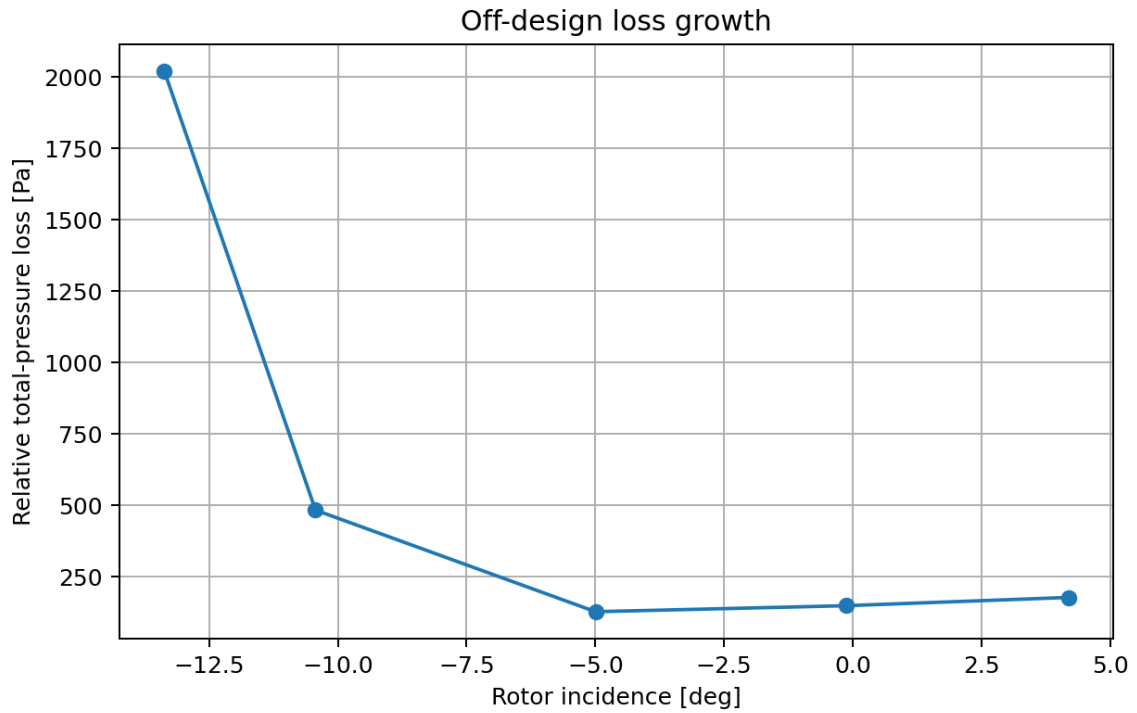


Figure 5: Total-pressure-loss quantities across the design and off-design cases. For the rotor, the loss is evaluated consistently in the relative frame.

8 Pressure losses

For incompressible flow, the kinematic total pressure is

$$p_t = p + \frac{1}{2}|\mathbf{U}|^2. \quad (18)$$

The physical total-pressure difference is obtained by multiplying by density,

$$\Delta p_{t,\text{phys}} = \rho (p_{t,\text{in}} - p_{t,\text{out}}). \quad (19)$$

For the rotor, $\mathbf{U} = \mathbf{W}$ in the CFD field, so rotor loss is evaluated in the relative frame. The reconstructed absolute velocity is used for velocity triangles and visualization, but it is not mixed into the relative-frame loss definition.

This distinction is essential. Mixing W at one station with V at another would create a frame-dependent quantity with no consistent physical meaning.

9 Baseline sequential visualization

For presentation purposes, the one-passage solutions are repeated in the tangential direction:

$$y_n = y_0 + n s, \quad n = -4, \dots, +4.$$

This creates a visual cascade without increasing the CFD cost.

For the rotor, the displayed absolute velocity is reconstructed as

$$\mathbf{V}_{\text{abs}} = \mathbf{W} + \mathbf{U}_b.$$

The stator already uses the absolute velocity.

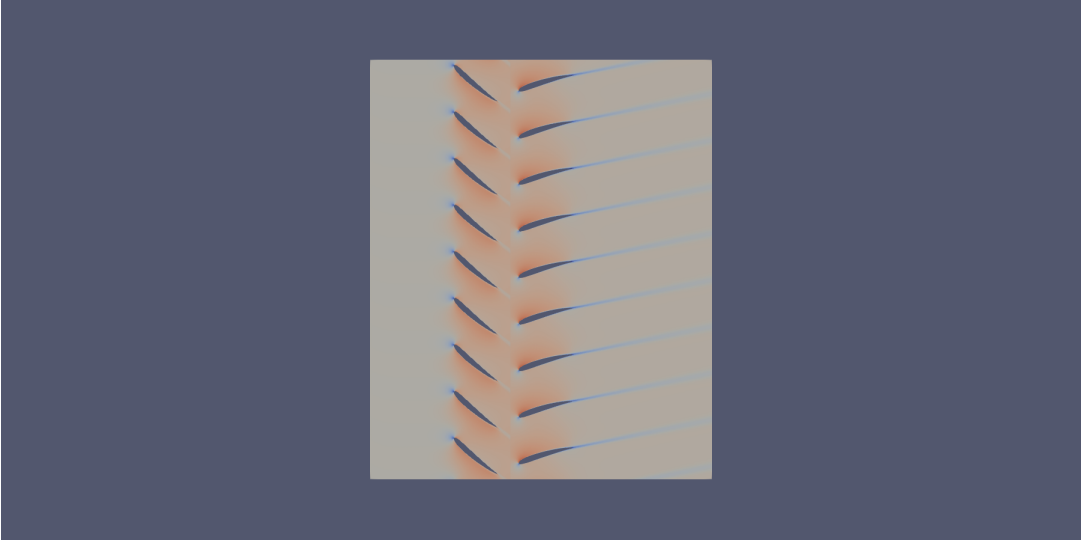


Figure 6: Composite rotor–stator visualization obtained by periodically replicating the independently solved rotor and stator passages. The vertical interface is a post-processing cut between two separate CFD solutions; it must not be interpreted as a physically coupled rotor–stator interface.

The discontinuity visible at the joining plane is therefore expected. It is not a numerical shock and not a physical prediction. It is a consequence of placing two separately solved fields next to one another for visualization.

This limitation is useful because it clearly motivates the next modelling step: a true coupled stage in which rotor and stator exchange pressure and velocity information during the same solution.

10 Current quantitative status

The quantities already verified from the executed nominal rotor case are summarized below.

Table 4: Known numerical quantities from the executed baseline project.

Quantity	Value
Rotor nominal cell count	87 053
Maximum reported skewness	2.125
Periodic-top faces	412
Periodic-bottom faces	412
Rotor inlet absolute velocity	(60, 0, 0) m/s
Rotor inlet relative velocity	(60, −80, 0) m/s
Blade speed	80 m/s
Relative inlet angle	−53.13°
Rotor LE metal angle	−53°
Design incidence	approximately −0.13°
Inlet k	24 m ² /s ²
Inlet ω	2.13×10^3 /s

The final rotor/stator outlet values are generated automatically in `results/summary.csv`. For a final portfolio version, the following values should be copied into the macros at the beginning of this file:

Table 5: Project result values to populate from the generated CSV file.

Quantity	Current document value
Rotor absolute exit angle α_2	30.10°
Stator absolute exit angle α_3	11.35°
Rotor exit tangential velocity $V_{\theta 2}$	34.78 m/s
Stator exit tangential velocity $V_{\theta 3}$	12.05 m/s
Mass-flow imbalance	3.3×10^{-8} %

11 Coupled rotor–stator MRF stage

The sequential cascade model was extended to a single-domain rotor–stator calculation. The purpose of this section is to document the numerical model and the post-processing quantities. Interpretation of the final coupled-stage results is intentionally deferred until the exported CSV files and the final PDF have been reviewed together.

11.1 Thin-annulus geometry and periodicity

The coupled domain is a quasi-two-dimensional annular sector centred on the machine axis x . The mean radius and radial thickness are

$$r_{\text{mean}} = 0.200 \text{ m}, \quad \Delta r = 1.5 \text{ mm}.$$

The same blade-to-blade pitch is used for rotor and stator,

$$s = 50 \text{ mm},$$

with one rotor blade and one stator blade per periodic sector. The axial gap between the rotor trailing edge and stator leading edge is approximately 15 mm.

The original unwrapped blade-to-blade coordinates (x, s, r) are mapped to the cylindrical sector according to

$$(x, s, r) \mapsto \left(x, r \sin \left(\frac{s}{r_{\text{mean}}} \right), r \cos \left(\frac{s}{r_{\text{mean}}} \right) \right).$$

The corresponding CFD sector angle is

$$\Delta\theta = \frac{s}{r_{\text{mean}}} = 0.25 \text{ rad} = 14.3239^\circ.$$

The two circumferential boundaries are conformal `cyclic` patches. Periodicity is imposed in Gmsh before the cylindrical mapping. No AMI is used: the rotor and stator belong to one conformal mesh and exchange pressure and velocity through ordinary internal faces.

11.2 Coupled mesh

The coupled mesh uses prismatic cells in the core and hexahedral layers near the blade walls. The final reference mesh uses three cells across the thin-annulus radial direction. The boundary-layer settings are approximately

$$\Delta y_1 \simeq 5 \mu\text{m}, \quad g \simeq 1.15, \quad N_{\text{BL}} \simeq 25,$$

with a total boundary-layer thickness of approximately 1.06 mm. Rounded trailing edges are used to avoid the boundary-layer fan generated by the earlier blunt trailing-edge construction.

Table 6: Reference coupled-stage mesh quantities.

Quantity	Value
Cells	186 240
Rotor-zone cells	89 304
Approximate maximum aspect ratio	95
Maximum non-orthogonality	54.9°
Mean non-orthogonality	2.25°
Maximum skewness	1.29
Minimum determinant	9.5×10^{-6}
Cells with determinant below 10^{-3}	33 442

The standard `checkMesh` checks are retained in the workflow. The small-determinant population is reported explicitly rather than omitted from the mesh description.

11.3 MRF formulation

The rotor is defined by the cell zone `rotorZone`. The stationary stator remains outside this zone. The MRF parameters are

$$\boldsymbol{\Omega} = (-400, 0, 0) \text{ rad/s},$$

with origin $(0, 0, 0)$ and rotation axis $(1, 0, 0)$. At the mean radius the peripheral speed magnitude is therefore

$$U_b = |\boldsymbol{\Omega}| r_{\text{mean}} = 80 \text{ m/s}.$$

The ordinary stationary patches intersecting the MRF zone are excluded from the rotating-patch correction through

```
nonRotatingPatches (inlet hub shroud);
```

while the rotor blade remains part of the MRF treatment.
The design absolute inlet velocity is

$$\mathbf{V}_1 = (60, 0, 0) \text{ m/s},$$

and the corresponding rotor-relative inlet velocity at the mean radius is

$$\mathbf{W}_1 = \mathbf{V}_1 - \mathbf{U}_b = (60, -80, 0) \text{ m/s},$$

so that $|\mathbf{W}_1| = 100 \text{ m/s}$ and $\beta_1 \simeq -53.13^\circ$.

11.4 Numerical setup and start-up procedure

The coupled model uses `simpleFoam`. The robust start-up discretization uses bounded first-order convection for velocity and turbulence, limited gradients and limited non-orthogonal diffusion:

- `gradSchemes: cellLimited Gauss linear 1;`
- `div(phi,U): bounded Gauss upwind;`
- `div(phi,k): bounded Gauss upwind;`
- `div(phi,omega): bounded Gauss upwind;`
- Laplacian interpolation: `Gauss linear limited 0.33;`
- surface-normal gradient: `limited 0.33;`
- two non-orthogonal pressure correctors.

Typical under-relaxation values used for the robust coupled calculation are

$$\alpha_p = 0.10\text{--}0.15, \quad \alpha_U = 0.15\text{--}0.25,$$

with lower relaxation for k and ω during SST activation.

The start-up procedure separates initialization from the final turbulent calculation. A laminar MRF solution is first established and the inlet velocity can be increased progressively before activating the k – ω SST model from the latest converged field. The final SST initialization is kept deliberately mild rather than scaling k directly with the 60 m/s inlet velocity.

The final `controlDict` supplied with the current case uses

Table 7: Current `controlDict` settings for the converged coupled case.

Setting	Value
Application	<code>simpleFoam</code>
<code>startFrom</code>	<code>latestTime</code>
<code>endTime</code>	3500
<code>writeInterval</code>	10
<code>purgeWrite</code>	4
Residual monitoring	<code>solverInfo: U, p, k, ω</code>
Field extrema	<code>fieldMinMax: U, p, k, ω, ν_t</code>
Flux monitoring	inlet and outlet sums of ϕ

11.5 Coupled-stage post-processing definitions

All quantitative post-processing is performed on the original one-sector CFD solution. The full-annulus reconstruction shown later is only a visualization of periodically repeated data.

For a quantity f , station averages are evaluated with mass-flux weighting,

$$\langle f \rangle_{\dot{m}} = \frac{\int_A \rho V_x f \, dA}{\int_A \rho V_x \, dA}. \quad (20)$$

For the cylindrical mapping used here,

$$r = \sqrt{y^2 + z^2}, \quad \mathbf{e}_\theta = \left(0, \frac{z}{r}, -\frac{y}{r}\right),$$

so that the absolute circumferential component is

$$V_\theta = \mathbf{V} \cdot \mathbf{e}_\theta. \quad (21)$$

The relative velocity in the rotor is reconstructed from the MRF solution as

$$\mathbf{W} = \mathbf{V} - \boldsymbol{\Omega} \times \mathbf{r}. \quad (22)$$

The absolute and relative flow angles are then

$$\alpha = \text{atan2}(V_\theta, V_x), \quad \beta = \text{atan2}(W_\theta, W_x). \quad (23)$$

For incompressible flow, the physical absolute total pressure used in the station post-processing is

$$p_{t,\text{abs}} = \rho \left(p_{\text{OF}} + \frac{1}{2} |\mathbf{V}|^2 \right), \quad (24)$$

and the rotor-relative total pressure is defined analogously with $|\mathbf{W}|$.

The current Python post-processing exports the following files:

```
results/postprocess/station_summary.csv
results/postprocess/stage_kpis.csv
results/postprocess/postprocess_report.txt
results/postprocess/convergence.csv
results/postprocess/wake_profiles/*.csv
```

No y^+ result is included in the present report version.

11.6 Full-annulus visualization

For visualization, the periodic sector is duplicated around the machine axis. The engineering quantities remain those of the original sector; the full-annulus reconstruction does not introduce additional independent flow solutions.

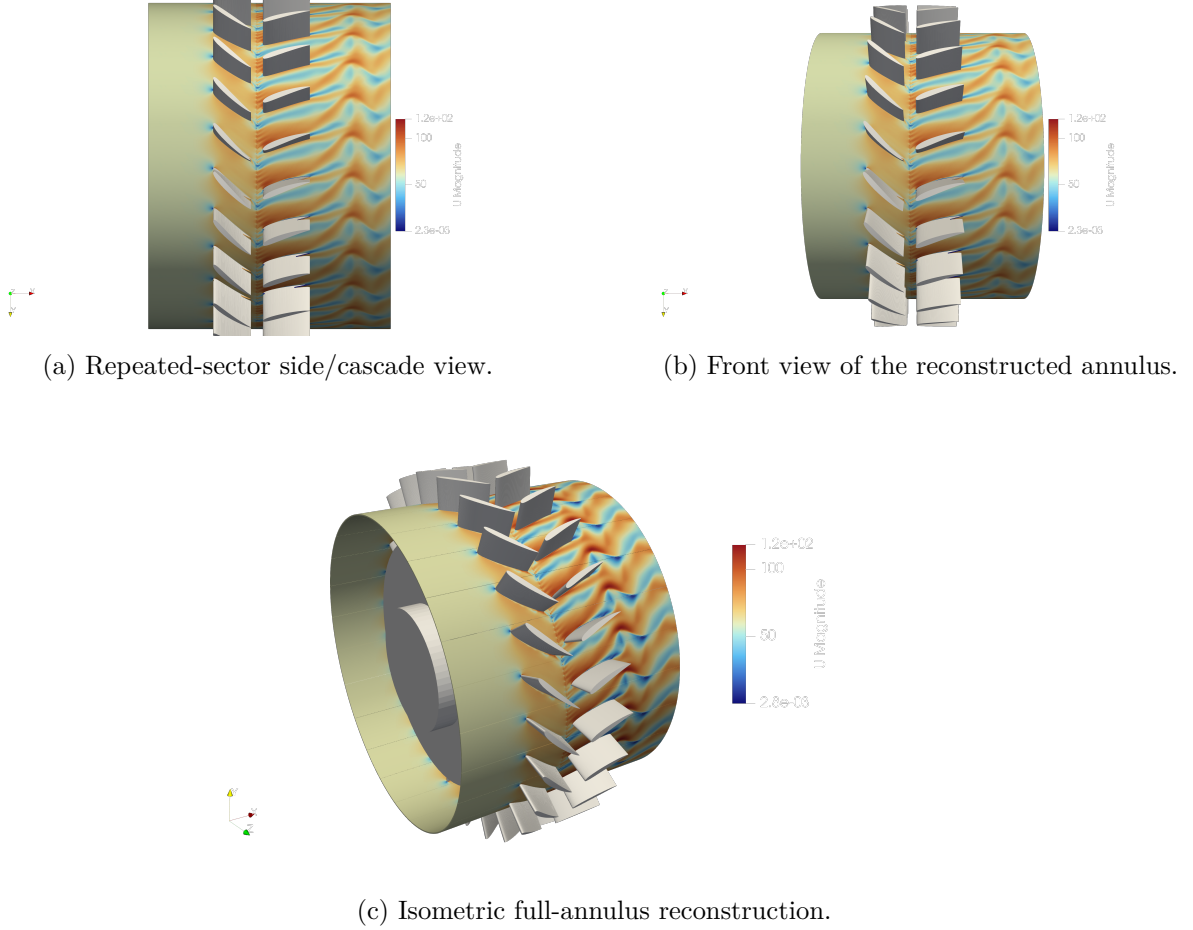


Figure 7: Velocity-magnitude visualization of the stable coupled rotor–stator MRF solution. The annular view is a periodic reconstruction of the one-sector result and is used for visualization only; all numerical quantities are extracted from the original CFD sector.

11.7 Coupled-stage results and discussion

The coupled calculation reaches the intended design operating point. At the rotor inlet the mass-flux-weighted axial velocity is 60.18 m/s, while the reconstructed relative flow angle is $\beta_1 = -53.063^\circ$. With a rotor leading-edge metal angle of approximately -53° , the resulting incidence is

$$i_R \simeq -0.063^\circ,$$

and the flow coefficient is

$$\phi = \frac{V_x}{U_b} = 0.752,$$

very close to the design value $60/80 = 0.75$. This agreement shows that the velocity triangle, blade speed and rotor geometry are mutually consistent in the coupled MRF calculation.

Rotor turning and comparison with the sequential baseline

The first post-processing script originally sampled “rotor exit” at $x = 5$ mm. However, the `rotorZone` generated by `topoSet` extends to $x = 7.5$ mm. That first sample is therefore still inside the MRF zone and should not be used as the final post-rotor station. The first available station outside the MRF region is at $x = 10$ mm, still upstream of the stator leading edge at approximately $x = 15$ mm. At this station the coupled solution gives

$$V_{\theta 2} = 36.03 \text{ m/s}, \quad \alpha_2 = 29.21^\circ.$$

The original sequential rotor calculation gave $V_{\theta 2} = 34.78 \text{ m/s}$ and $\alpha_2 = 30.10^\circ$. Thus the more complete coupled model retains almost the same rotor turning: the absolute exit angle changes by less than one degree and the tangential velocity by only about 1.25 m/s . This provides a useful consistency check between the reduced cascade model and the MRF extension.

Using the first station outside `rotorZone`, the Euler estimate is

$$\Delta h_{0,\text{Euler}} = U_b (V_{\theta 2} - V_{\theta 1}) \approx 2.89 \text{ kJ/kg},$$

corresponding to a loading coefficient

$$\psi = \frac{\Delta h_0}{U_b^2} \approx 0.451.$$

These corrected values replace the preliminary post-processing values $\Delta h_0 = 2.10 \text{ kJ/kg}$ and $\psi = 0.329$, which were obtained from the sample located inside the MRF region.

Stator straightening

The stator receives a mean absolute inlet angle of $\alpha_{S,\text{in}} = 29.21^\circ$. The stator leading-edge metal angle is approximately 28.1° , giving a coupled-stage incidence of only

$$i_S \simeq 1.11^\circ.$$

At the stator exit the mean tangential component and absolute flow angle are

$$V_{\theta 3} = 17.96 \text{ m/s}, \quad \alpha_3 = 15.95^\circ.$$

Hence the stator removes approximately 50.2 % of the post-rotor tangential velocity and 45.4 % of the flow angle before the nominal stator-exit station. Farther downstream, at the outlet, the corresponding reductions reach approximately 57.6 % and 52.9 %. The stator therefore performs the intended straightening, but less strongly than in the sequential calculation, where the angular straightening was about 62.3 %.

This difference is physically meaningful. In the sequential model the stator was supplied with a uniform inlet condition reconstructed from the mean rotor exit. In the coupled model it receives the actual non-uniform rotor wake and pressure field. The reduced straightening therefore quantifies one of the main effects that the one-way workflow could not represent.

Table 8: Sequential and coupled-stage turning comparison.

Quantity	Sequential baseline	Coupled MRF
Rotor-exit α_2	30.10°	29.21°
Rotor-exit $V_{\theta 2}$	34.78 m/s	36.03 m/s
Stator-exit α_3	11.35°	15.95°
Stator-exit $V_{\theta 3}$	12.05 m/s	17.96 m/s
Angular straightening	62.3 %	45.4 %
V_θ straightening	65.4 %	50.2 %

Wake transport and mixing

The circumferential profiles show why the coupled result differs from the uniform-inlet sequential model. Immediately upstream of the stator the velocity magnitude ranges from approximately 31.6 m/s to 96.3 m/s over one pitch. Downstream of the stator this range narrows to approximately 36.0 m/s – 82.0 m/s , and at the outlet to approximately 45.4 m/s – 73.0 m/s . The coupled model therefore resolves a strong rotor wake entering the stator and its progressive mixing downstream rather than replacing it by a mean velocity.

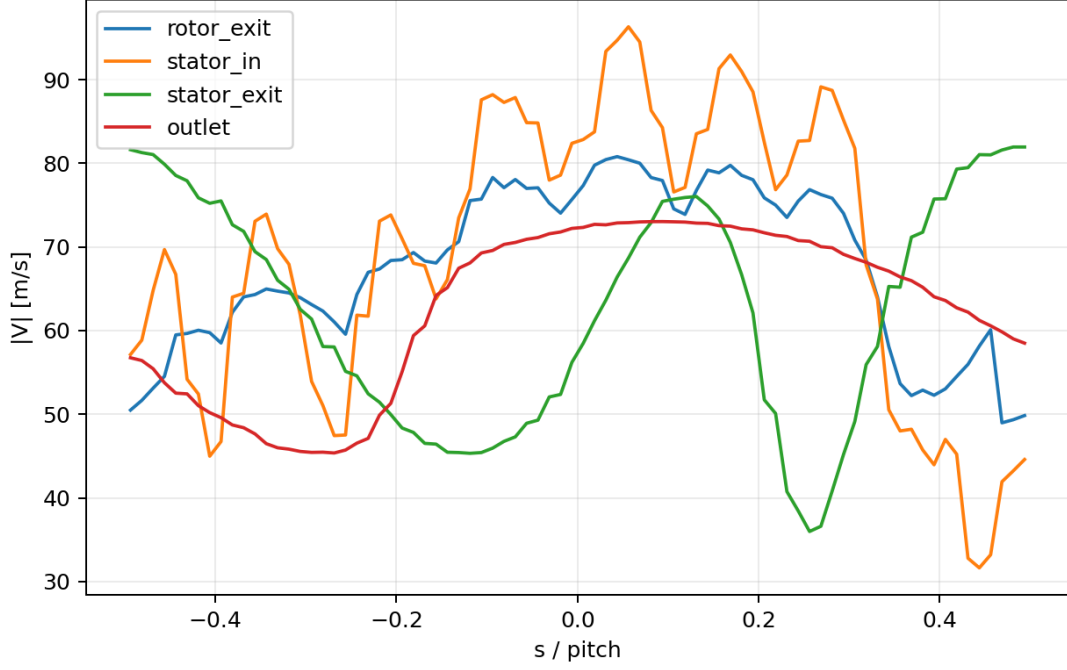


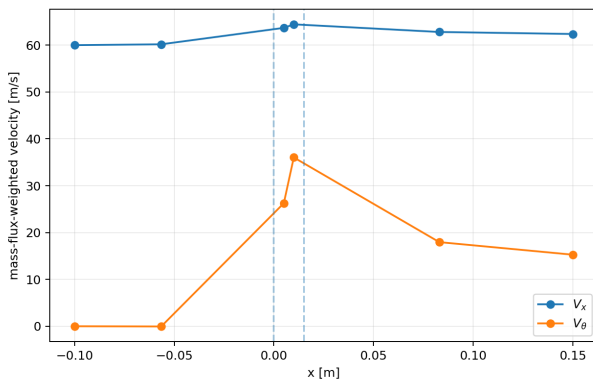
Figure 8: Circumferential velocity-magnitude profiles at the coupled-stage sampling stations. The strong non-uniformity upstream of the stator is progressively reduced downstream.

Mean velocity, angle and pressure evolution

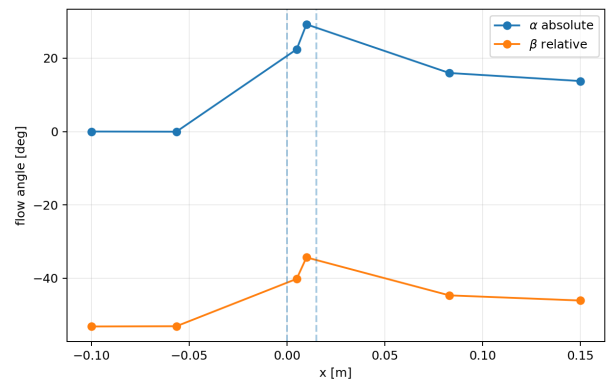
The mass-flux-weighted station data show that the rotor generates positive swirl while the stator subsequently removes a substantial fraction of it. The static pressure rises from approximately -3.99 kPa at the inlet to -0.18 kPa at the outlet, corresponding to

$$\Delta p_{\text{static}} \approx 3.81 \text{ kPa}.$$

Only pressure differences are physically meaningful because the incompressible pressure reference is arbitrary.



(a) Mass-flux-weighted velocity components.



(b) Absolute and relative flow angles.

Figure 9: Evolution of the mean flow through the coupled rotor–stator stage.

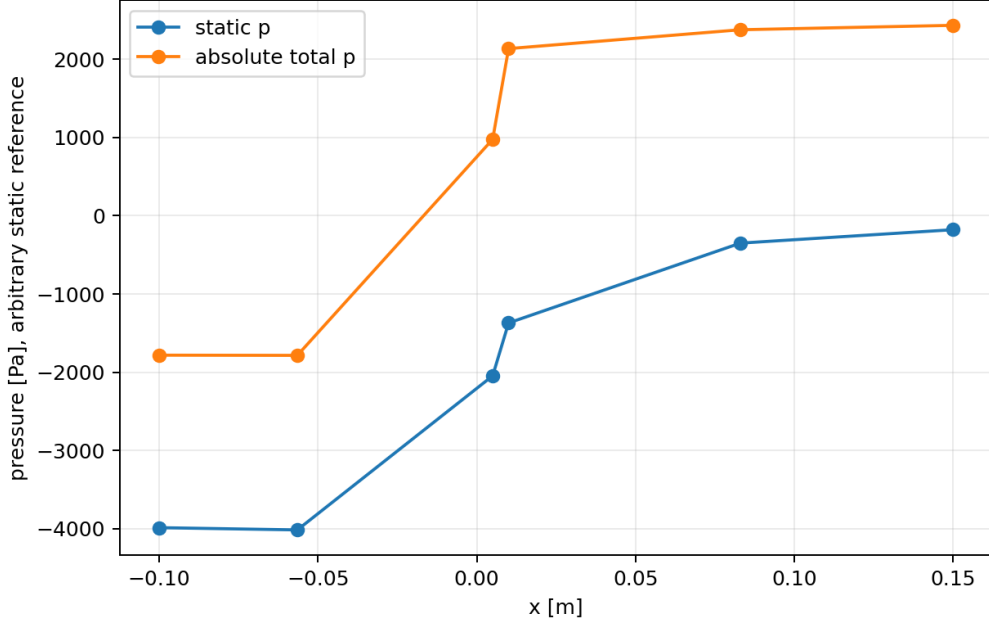


Figure 10: Mass-flux-weighted static and absolute-total-pressure diagnostics at the axial sampling stations.

Mass conservation and convergence

The patch-integrated volumetric fluxes at the final iteration are

$$Q_{\text{in}} = -4.499973384 \times 10^{-3} \text{ m}^3/\text{s}, \quad Q_{\text{out}} = +4.499973249 \times 10^{-3} \text{ m}^3/\text{s},$$

which corresponds to a relative imbalance of only

$$\epsilon_Q \approx 3.0 \times 10^{-6} \text{ \%}.$$

The larger 0.0188 % imbalance reported by the ParaView sampling script is therefore mainly an interpolation/integration error of the cut surfaces rather than a global continuity defect.

The final solution is best described as *stable and quasi-converged* rather than mathematically fully converged. At iteration 3500 the initial velocity residuals remain around 9×10^{-4} – 1.1×10^{-3} , the k residual is about 3.1×10^{-4} , while the ω residual has fallen below 10^{-6} . The global mass balance is nevertheless excellent and the integrated flow quantities have reached a stable operating state. Local extrema still exhibit small variations near the end of the run, so conclusions based on single-cell maxima should be treated more cautiously than the mass-averaged station quantities.

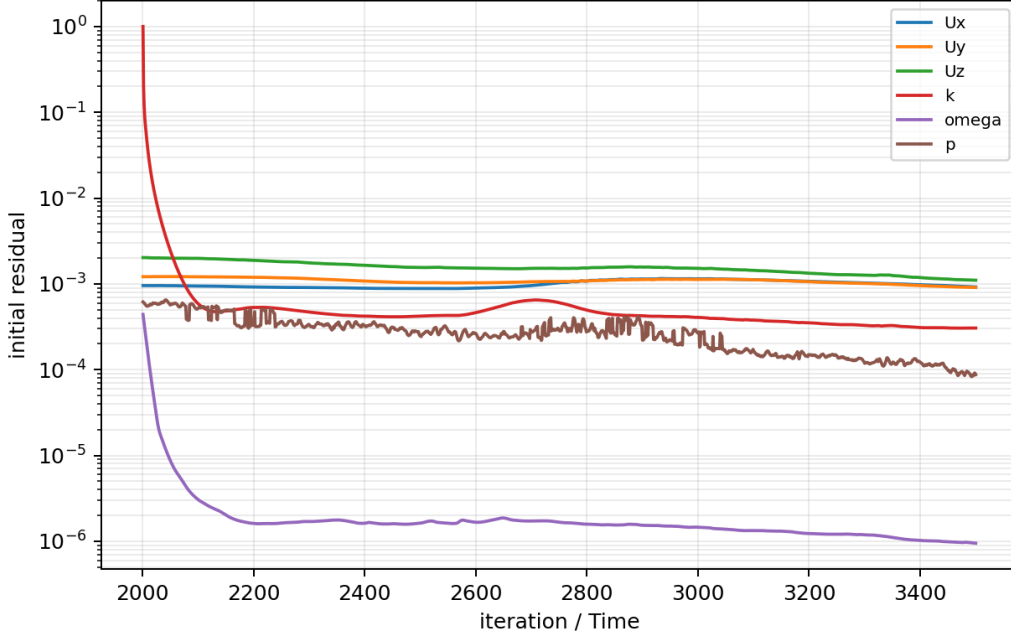


Figure 11: Residual history of the final coupled SST calculation. The velocity equations reach a residual plateau of order 10^{-3} while the global mass balance remains very small.

MRF-interface sensitivity

A notable numerical feature is located at the downstream edge of `rotorZone`, $x = 7.5$ mm. At the final iteration the maximum velocity magnitude is

$$|\mathbf{U}|_{\max} = 120.9 \text{ m/s}$$

at $x = 7.437$ mm, only about 0.063 mm before the MRF-zone boundary. The maximum pressure occurs at $x = 7.847$ mm, about 0.35 mm downstream of the same boundary. The visually sharp axial seam seen in the velocity plots therefore coincides with real local extrema and should not be dismissed as a pure rendering artefact.

This observation does not by itself invalidate the MRF calculation, because the rotating-frame source term necessarily terminates at the edge of the cell zone. It does, however, make this interface the first location for a targeted validation. Additional cuts immediately upstream and downstream of $x = 7.5$ mm should be used to determine whether V_x , V_θ , p and p_t remain continuous or contain a non-physical jump.

Total-pressure and energy-balance caution

The preliminary script reports an absolute-total-pressure rise of about 4.22 kPa. It also reports negative “losses” across both the rotor and stator and an Euler-to-total-pressure proxy efficiency greater than unity. Those quantities are not physically acceptable as loss or efficiency metrics in their present form: a passive stator cannot generate total pressure and an efficiency above unity signals an inconsistent station/frame definition.

Part of the inconsistency comes from the original rotor-exit sample being inside `rotorZone`. Replacing it by the first post-MRF station changes the Euler work estimate from about 2.10 kJ/kg to about 2.89 kJ/kg, but the total-pressure balance still requires a frame-consistent verification. Consequently, the current report uses the static-pressure rise, turning, wake transport and mass balance as validated quantities, while the total-pressure-based efficiency and row-loss values are retained only as diagnostics. A useful next check is an independent torque balance on the rotor blade,

$$P = \Omega T_x, \quad w = \frac{P}{\dot{m}},$$

which can be compared directly with $U_b \Delta V_\theta$.

Turbulence field and near-wall interpretation

The mass-averaged turbulent kinetic energy increases from approximately $0.005 \text{ m}^2/\text{s}^2$ at the inlet to $3.07 \text{ m}^2/\text{s}^2$ at the first rotor-exit sample and $4.98 \text{ m}^2/\text{s}^2$ just upstream of the stator, before decreasing to $0.61 \text{ m}^2/\text{s}^2$ at the stator exit and $0.009 \text{ m}^2/\text{s}^2$ at the outlet. This is consistent with turbulence production in the rotor/wake region followed by downstream mixing and dissipation.

The final local maximum is much larger, $k_{\max} \approx 213 \text{ m}^2/\text{s}^2$, in the rotor region. This hotspot should be examined spatially before being given a physical interpretation. The maximum turbulent viscosity is $\nu_{t,\max} \approx 3.11 \times 10^{-3} \text{ m}^2/\text{s}$, about 207 times the molecular kinematic viscosity. The wall value $\omega_{\max} \approx 1.93 \times 10^8 \text{ 1/s}$ is not, on its own, evidence of divergence: with a first-cell height of order $5 \mu\text{m}$, the `omegaWallFunction` can naturally produce very large near-wall values. A definitive wall-resolution assessment still requires the missing y^+ post-processing.

Mesh quality and incompressible-model limits

The conventional mesh metrics remain acceptable for this reduced model, with maximum non-orthogonality of 54.9° and maximum skewness of 1.29. However, 33,442 of 186,240 cells, about 18 %, have a determinant below 10^{-3} . This does not mean that the same fraction of the mesh is invalid, because the thin-annulus and boundary-layer construction naturally creates highly anisotropic cells, but it does mean that isolated extrema should be interpreted with more caution than integral quantities.

The final maximum absolute velocity is about 120.9 m/s . With a reference speed of sound of 343 m/s , this corresponds to $M \approx 0.35$. The incompressible model therefore remains useful for the present reduced-order study, but local velocities have entered a range where a compressible formulation would be preferable for accurate pressure predictions.

Table 9: Principal coupled-stage results retained for interpretation.

Quantity	Value
Rotor incidence	-0.063°
Flow coefficient ϕ	0.752
Post-rotor α_2	29.21°
Post-rotor $V_{\theta 2}$	36.03 m/s
Stator-exit α_3	15.95°
Stator-exit $V_{\theta 3}$	17.96 m/s
Stator V_θ straightening	50.2 %
Corrected Euler-work estimate	2.89 kJ/kg
Corrected loading coefficient ψ	0.451
Static-pressure rise, inlet to outlet	3.81 kPa
Patch-flux mass imbalance	$3.0 \times 10^{-6} \%$
Maximum absolute velocity	120.9 m/s

12 Limitations of the coupled model

The coupled model remains deliberately reduced in scope:

- only one rotor passage and one stator passage are solved, with circumferential periodicity;

- the domain is a thin annulus around a fixed mean radius, so spanwise blade design, realistic hub-to-shroud variation and full radial equilibrium are not represented;
- tip clearance is absent;
- the rotor is represented through a steady MRF/frozen-rotor formulation rather than a transient sliding mesh, so blade-passing unsteadiness is absent;
- the final SST solution is stable but retains a velocity-residual plateau of order 10^{-3} and is therefore described as quasi-converged rather than perfectly converged;
- the MRF-zone downstream boundary at $x = 7.5$ mm coincides with strong local velocity and pressure extrema and should be checked with additional interface-normal cuts;
- total-pressure-based row losses and the preliminary efficiency proxy are not retained as validated results until a frame-consistent energy balance or independent rotor-torque calculation is completed;
- the flow is modelled as incompressible although local absolute velocity reaches about 121 m/s, corresponding to approximately $M = 0.35$ for $a = 343$ m/s;
- the turbulence model is steady RANS $k-\omega$ SST;
- approximately 18 % of the final cells have determinant below 10^{-3} , so local single-cell extrema should not be over-interpreted;
- the present report does not yet include a y^+ verification, so the intended wall-resolved near-wall treatment has not yet been confirmed quantitatively.

13 Next numerical extension

The most useful next steps are ordered by validation value rather than by model complexity. First, the MRF-interface region should be sampled with several closely spaced axial cuts around $x = 7.5$ mm and the rotor blade torque should be integrated to close the energy balance against $U_b \Delta V_\theta$. Second, y^+ and wall-shear-stress fields should be computed on rotor and stator to confirm the near-wall resolution and identify any local separation. Third, a small coupled-stage operating sweep in V_x at fixed Ω would provide stage-level trends in turning, pressure rise and wake behaviour.

Only after those checks would a transient sliding-interface calculation be the natural next modelling step. It would resolve blade-passing effects and the time-dependent relative motion of rotor wakes and stator blades. A compressible formulation is also justified if quantitative pressure prediction is required at the observed local Mach numbers.

14 Conclusion

The project progresses from an independently solved rotor and stator cascade to a genuinely coupled thin-annulus MRF stage. The coupled calculation retains the intended design point: the rotor inlet incidence is only -0.063° and the flow coefficient is $\phi = 0.752$. The rotor generates approximately 36.03 m/s of absolute swirl and a post-rotor flow angle of 29.21° , closely reproducing the 30.10° rotor-exit angle obtained with the earlier sequential model. This agreement provides a useful cross-check between the two modelling levels.

The coupled stator reduces the tangential velocity to 17.96 m/s and the mean absolute flow angle to 15.95° . Its straightening is weaker than in the sequential model because the stator now receives the strongly non-uniform rotor wake rather than a uniform mean inlet. The wake profiles show a velocity-magnitude range of roughly 32 m/s– 96 m/s just upstream of the stator, followed by progressive mixing downstream. Capturing this direct wake–stator interaction is the main physical gain of the coupled model.

The calculation also produces a clear static-pressure rise of approximately 3.81 kPa and excellent patch-integrated mass conservation, with a final inlet/outlet flux mismatch of only $3.0 \times 10^{-6} \%$. The solution is nevertheless described as quasi-converged rather than perfectly converged, because the velocity residuals plateau near 10^{-3} and local extrema retain small late-iteration variations.

Two issues remain open before the model should be used for stronger performance claims. First, the downstream edge of `rotorZone` is coincident with strong local extrema, so the MRF interface requires a targeted continuity check. Second, the preliminary total-pressure loss and efficiency metrics are not energetically self-consistent and are therefore not used as validated performance quantities. A direct rotor-torque calculation and a corrected station definition are the appropriate next checks. In addition, the local absolute velocity reaches about 120.9 m/s ($M \approx 0.35$), which marks the onset of a regime where compressibility should be considered for quantitative pressure prediction.

Within these limits, the project now demonstrates more than textbook velocity triangles: it links parametric blade geometry, periodic meshing, robust MRF start-up, steady RANS, two-way rotor–stator coupling, wake transport, turning, straightening and quantitative conservation checks in one reproducible OpenFOAM workflow.